

Feature Selection in Statistical Machine Learning

Garvesh Raskutti

Lecture 1

Plan for today

- Logistics, assessment, e.t.c.
- Motivation: What and why?
- Simple example

Plan for today

- Logistics, assessment, e.t.c.
- Motivation: What and why?
- Simple example

Plan for today

- Logistics, assessment, e.t.c.
- Motivation: What and why?
- Simple example

Syllabus

- Office hours
- Pre-requisites
- Assessment

Office hours

- TA Chenghui Zheng: Tuesday 10-11AM, MH 2517
- Professor: Thursday 9-10AM, MH 6659

Pre-requisites

- Stat 333 or 340 (basic probability and statistical knowledge)
- Math 222 (basic linear algebra)
- Any more challenging math will be introduced in this course

Pre-requisites

- Stat 333 or 340 (basic probability and statistical knowledge)
- Math 222 (basic linear algebra)
- Any more challenging math will be introduced in this course

Pre-requisites

- Stat 333 or 340 (basic probability and statistical knowledge)
- Math 222 (basic linear algebra)
- Any more challenging math will be introduced in this course

Key assumed concepts

- Random variables
- Conditional probability
- Expectation and conditional expectation
- Linear model and linear regression
- Correlation

Textbook/references

- No single textbook but references provided in syllabus
- For each week, topics are listed with reference to the textbook
- Lectures will be entirely self-contained
- Sparse lecture notes (along with discussion and whiteboard explanations) will be provided

Assessment overview

- Attendance and participation (10%)
- Homework assignments (20%) (generative AI allowed)
- Project (30%) (generative AI allowed)
- Exam (40%) (generative AI NOT allowed)

Assessment overview

- Attendance and participation (10%)
- Homework assignments (20%) (generative AI allowed)
- Project (30%) (generative AI allowed)
- Exam (40%) (generative AI NOT allowed)

Assessment overview

- Attendance and participation (10%)
- Homework assignments (20%) (generative AI allowed)
- Project (30%) (generative AI allowed)
- Exam (40%) (generative AI NOT allowed)

Assessment overview

- Attendance and participation (10%)
- Homework assignments (20%) (generative AI allowed)
- Project (30%) (generative AI allowed)
- Exam (40%) (generative AI NOT allowed)

Attendance and participation

- Attendance at lectures is highly encouraged since all the assessment will be based on the material covered in lectures.
- Since critical thinking will be what is assessed most, the lectures will attempt to put students in the best position to perform the critical thinking to do well.
- How that 10% is determined is provided in the syllabus.

Attendance and participation

- Attendance at lectures is highly encouraged since all the assessment will be based on the material covered in lectures.
- Since critical thinking will be what is assessed most, the lectures will attempt to put students in the best position to perform the critical thinking to do well.
- How that 10% is determined is provided in the syllabus.

Attendance and participation

- Attendance at lectures is highly encouraged since all the assessment will be based on the material covered in lectures.
- Since critical thinking will be what is assessed most, the lectures will attempt to put students in the best position to perform the critical thinking to do well.
- How that 10% is determined is provided in the syllabus.

Homework assignments

- There will be 4 assignments worth a total of 20%
- Due dates: Feb. 19, Mar. 12, Apr. 2, Apr. 23.
- Zero tolerance policy on late homework
- Generative AI use permitted but strongly recommended not to copy and paste directly

Homework assignments

- There will be 4 assignments worth a total of 20%
- Due dates: Feb. 19, Mar. 12, Apr. 2, Apr. 23.
- Zero tolerance policy on late homework
- Generative AI use permitted but strongly recommended not to copy and paste directly

Homework assignments

- There will be 4 assignments worth a total of 20%
- Due dates: Feb. 19, Mar. 12, Apr. 2, Apr. 23.
- Zero tolerance policy on late homework
- Generative AI use permitted but strongly recommended not to copy and paste directly

Homework assignments

- There will be 4 assignments worth a total of 20%
- Due dates: Feb. 19, Mar. 12, Apr. 2, Apr. 23.
- Zero tolerance policy on late homework
- Generative AI use permitted but strongly recommended not to copy and paste directly

Project

- Course project worth 30%
- See course website for outline
- I'll discuss the course project in greater detail later (some time in February)

Exam

- To be given near the end of semester (last allotted lecture)
- Based on material covered in the course
- There will be no technology usage allowed

General tips

- Assessment will include a combination of math, coding, and most importantly critical thinking
- For example, interpreting results, understanding why one method performs better or worse compared to another method and suggesting alternative directions
- The best way to prepare is to attend lecture and have a solid understanding of the framework and methods and how they apply to different case studies presented in lecture

General tips

- Assessment will include a combination of math, coding, and most importantly critical thinking
- For example, interpreting results, understanding why one method performs better or worse compared to another method and suggesting alternative directions
- The best way to prepare is to attend lecture and have a solid understanding of the framework and methods and how they apply to different case studies presented in lecture

General tips

- Assessment will include a combination of math, coding, and most importantly critical thinking
- For example, interpreting results, understanding why one method performs better or worse compared to another method and suggesting alternative directions
- The best way to prepare is to attend lecture and have a solid understanding of the framework and methods and how they apply to different case studies presented in lecture

What is feature?

- In a supervised learning setting (we will discuss the unsupervised setting shortly), *feature* typically refers to a measurable quantity that potentially has an impact on a response variable
- E.g. *square footage* or *number of bedrooms* may be features when the response is house price
- E.g. *family background* could be a feature for a *mental health outcome*
- E.g. *game time* is a feature for *points scored* in a basketball game
- In neural networks/LLMs features may be *learned* or *hidden*

What is feature?

- In a supervised learning setting (we will discuss the unsupervised setting shortly), *feature* typically refers to a measurable quantity that potentially has an impact on a response variable
- E.g. *square footage* or *number of bedrooms* may be features when the response is house price
- E.g. *family background* could be a feature for a *mental health outcome*
- E.g. *game time* is a feature for *points scored* in a basketball game
- In neural networks/LLMs features may be *learned* or *hidden*

What is feature?

- In a supervised learning setting (we will discuss the unsupervised setting shortly), *feature* typically refers to a measurable quantity that potentially has an impact on a response variable
- E.g. *square footage* or *number of bedrooms* may be features when the response is house price
- E.g. *family background* could be a feature for a *mental health outcome*
- E.g. *game time* is a feature for *points scored* in a basketball game
- In neural networks/LLMs features may be *learned* or *hidden*

What is feature?

- In a supervised learning setting (we will discuss the unsupervised setting shortly), *feature* typically refers to a measurable quantity that potentially has an impact on a response variable
- E.g. *square footage* or *number of bedrooms* may be features when the response is house price
- E.g. *family background* could be a feature for a *mental health outcome*
- E.g. *game time* is a feature for *points scored* in a basketball game
- In neural networks/LLMs features may be *learned* or *hidden*

What is feature?

- In a supervised learning setting (we will discuss the unsupervised setting shortly), *feature* typically refers to a measurable quantity that potentially has an impact on a response variable
- E.g. *square footage* or *number of bedrooms* may be features when the response is house price
- E.g. *family background* could be a feature for a *mental health outcome*
- E.g. *game time* is a feature for *points scored* in a basketball game
- In neural networks/LLMs features may be *learned* or *hidden*

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is a feature cont?

- The term *feature* is often used interchangeably with *variable* and *attribute*
- Distinctions are somewhat arbitrary and artificial
- Sometimes attributes and variables are considered to be broader than features since they are measurable quantities potentially before they are pre-processed and used in a machine learning algorithm
- Sometimes a feature is referred to as an *independent* variable whereas the response is referred to as the *dependent* variable
- For the purpose of this course, we will generally consider a feature to be a measurable quantity that potentially influences the response
- From a loose notation perspective, Y 's will refer to responses and X 's will be referred to as features

What is feature selection?

- **Goal:** Determine which features impact the response (will make precise later)
- Ideally feature set is small
- Distinct from feature engineering in which features are pre-processed to fit an algorithm
- In this course, we will assume features have all been engineered/pre-processed

What is feature selection?

- **Goal:** Determine which features impact the response (will make precise later)
- Ideally feature set is small
- Distinct from feature engineering in which features are pre-processed to fit an algorithm
- In this course, we will assume features have all been engineered/pre-processed

What is feature selection?

- **Goal:** Determine which features impact the response (will make precise later)
- Ideally feature set is small
- Distinct from feature engineering in which features are pre-processed to fit an algorithm
- In this course, we will assume features have all been engineered/pre-processed

What is feature selection?

- **Goal:** Determine which features impact the response (will make precise later)
- Ideally feature set is small
- Distinct from feature engineering in which features are pre-processed to fit an algorithm
- In this course, we will assume features have all been engineered/pre-processed

Motivation: Why learn about/study feature selection?

Motivation: Why learn about/study feature selection?

- (1) Domain significance, causality, interpretability, human understanding
- (2) Prediction performance, data quality, generalizability, robustness
- (3) Computational efficiency

Motivation: Why learn about/study feature selection?

- (1) Domain significance, causality, interpretability, human understanding
- (2) Prediction performance, data quality, generalizability, robustness
- (3) Computational efficiency

Motivation: Why learn about/study feature selection?

- (1) Domain significance, causality, interpretability, human understanding
- (2) Prediction performance, data quality, generalizability, robustness
- (3) Computational efficiency

Domain significance, causality, interpretability, human understanding

- From a human understanding perspective, feature selection is motivated in science by simplifying our understanding of models (i.e. Occam's razor)
- Connected to interpretable machine learning, causality, fairness, bias, e.t.c.
- Sometimes feature selection is a concrete scientific goal tied to desirable causal interventions
- **Example:** Systems biology where we want to determine which set of genes/proteins significant change a particular response (e.g. disease prevalence, energy gain, e.t.c.)
- By identifying a smaller subset of genes/proteins expensive interventions can be performed in a targeted way

Domain significance, causality, interpretability, human understanding

- From a human understanding perspective, feature selection is motivated in science by simplifying our understanding of models (i.e. Occam's razor)
- Connected to interpretable machine learning, causality, fairness, bias, e.t.c.
- Sometimes feature selection is a concrete scientific goal tied to desirable causal interventions
- **Example:** Systems biology where we want to determine which set of genes/proteins significant change a particular response (e.g. disease prevalence, energy gain, e.t.c.)
- By identifying a smaller subset of genes/proteins expensive interventions can be performed in a targeted way

Domain significance, causality, interpretability, human understanding

- From a human understanding perspective, feature selection is motivated in science by simplifying our understanding of models (i.e. Occam's razor)
- Connected to interpretable machine learning, causality, fairness, bias, e.t.c.
- Sometimes feature selection is a concrete scientific goal tied to desirable causal interventions
- **Example:** Systems biology where we want to determine which set of genes/proteins significant change a particular response (e.g. disease prevalence, energy gain, e.t.c.)
- By identifying a smaller subset of genes/proteins expensive interventions can be performed in a targeted way

Domain significance, causality, interpretability, human understanding

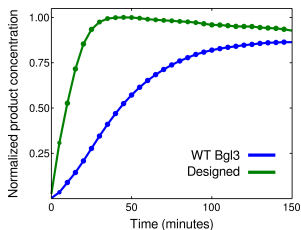
- From a human understanding perspective, feature selection is motivated in science by simplifying our understanding of models (i.e. Occam's razor)
- Connected to interpretable machine learning, causality, fairness, bias, e.t.c.
- Sometimes feature selection is a concrete scientific goal tied to desirable causal interventions
- **Example:** Systems biology where we want to determine which set of genes/proteins significant change a particular response (e.g. disease prevalence, energy gain, e.t.c.)
- By identifying a smaller subset of genes/proteins expensive interventions can be performed in a targeted way

Domain significance, causality, interpretability, human understanding

- From a human understanding perspective, feature selection is motivated in science by simplifying our understanding of models (i.e. Occam's razor)
- Connected to interpretable machine learning, causality, fairness, bias, e.t.c.
- Sometimes feature selection is a concrete scientific goal tied to desirable causal interventions
- **Example:** Systems biology where we want to determine which set of genes/proteins significant change a particular response (e.g. disease prevalence, energy gain, e.t.c.)
- By identifying a smaller subset of genes/proteins expensive interventions can be performed in a targeted way

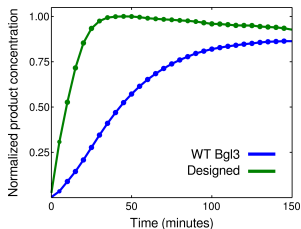
Example: Improving sugar production in photosynthesis by synthetically modifying a protein

- X data is protein sequences and Y is sugar/glucose production/yield
- Features are amino acid presences in different protein locations
 $p = 500^{20}$
- We developed a feature selection algorithm and selected 10 most significant features to intervene on



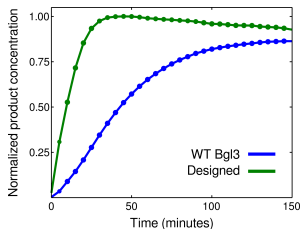
Example: Improving sugar production in photosynthesis by synthetically modifying a protein

- X data is protein sequences and Y is sugar/glucose production/yield
- Features are amino acid presences in different protein locations
 $p = 500^{20}$
- We developed a feature selection algorithm and selected 10 most significant features to intervene on



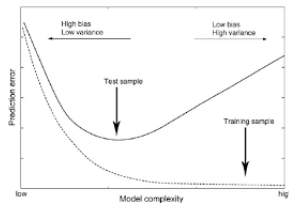
Example: Improving sugar production in photosynthesis by synthetically modifying a protein

- X data is protein sequences and Y is sugar/glucose production/yield
- Features are amino acid presences in different protein locations
 $p = 500^{20}$
- We developed a feature selection algorithm and selected 10 most significant features to intervene on



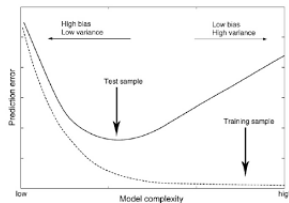
Prediction/statistical performance

- Including all features in a model does not hurt performance on training error
- Features that ultimately are not relevant can fit noise, overfit and lead to poor performance in a new sample (test error)
- Potentially due to: (i) overfitting; (ii) distribution shift (e.g. using models based on data from one race/gender to fit data from a different race/gender), and others
- Models with only relevant features can mitigate against this



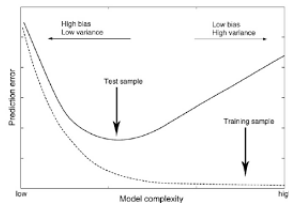
Prediction/statistical performance

- Including all features in a model does not hurt performance on training error
- Features that ultimately are not relevant can fit noise, overfit and lead to poor performance in a new sample (test error)
- Potentially due to: (i) overfitting; (ii) distribution shift (e.g. using models based on data from one race/gender to fit data from a different race/gender), and others
- Models with only relevant features can mitigate against this



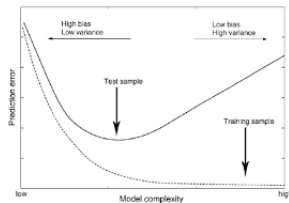
Prediction/statistical performance

- Including all features in a model does not hurt performance on training error
- Features that ultimately are not relevant can fit noise, overfit and lead to poor performance in a new sample (test error)
- Potentially due to: (i) overfitting; (ii) distribution shift (e.g. using models based on data from one race/gender to fit data from a different race/gender), and others
- Models with only relevant features can mitigate against this



Prediction/statistical performance

- Including all features in a model does not hurt performance on training error
- Features that ultimately are not relevant can fit noise, overfit and lead to poor performance in a new sample (test error)
- Potentially due to: (i) overfitting; (ii) distribution shift (e.g. using models based on data from one race/gender to fit data from a different race/gender), and others
- Models with only relevant features can mitigate against this



- A smaller number of features naturally leads to less storage and run-time
- Algorithms typically scale to some degree in terms of number selected features s both in storage and run-time
- For example, if we consider linear regression with n samples and s selected features, the storage scales as $O(ns)$ and the run-time scales as $O(ns \cdot \min(n, s))$

Computational efficiency

- A smaller number of features naturally leads to less storage and run-time
- Algorithms typically scale to some degree in terms of number selected features s both in storage and run-time
- For example, if we consider linear regression with n samples and s selected features, the storage scales as $O(ns)$ and the run-time scales as $O(ns \cdot \min(n, s))$

Computational efficiency

- A smaller number of features naturally leads to less storage and run-time
- Algorithms typically scale to some degree in terms of number selected features s both in storage and run-time
- For example, if we consider linear regression with n samples and s selected features, the storage scales as $O(ns)$ and the run-time scales as $O(ns \cdot \min(n, s))$

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Why take this course?

- While feature selection is widely viewed as useful for reasons outlined earlier, what can you expect to get out of this course?
- I.e. what can you get out of this course that hopefully goes beyond querying ChatGPT?
- There are also many standard packages that perform feature selection (e.g. Lasso, RFE, SelectKBest, e.t.c.)
- Focus of this course is on the understanding, critical thinking and fundamentals of feature selection
- It is most important (especially in job interviews and on the job market) is being able to interpret results from these packages and provide insight to bosses, customers, collaborators, e.t.c.
- E.g. Different packages/algorithms give you completely different results. How do you explain this to a stakeholder
- This course provides a framework to address interpretability and explainability questions like this
- Assessment also focuses on the interpretability and critical thinking rather than simply knowing and running packages

Example: Housing data

Consider a very simple example where we observe $n = 300$ house prices and there are $p = 3$ features: square footage, number of bedrooms, and a neighborhood quality score. Y : House price; X_1 : Square footage; X_2 : Number of bedrooms; X_3 : Neighborhood quality score. Below is a snapshot of the data.

id	price	sqft	bedrooms	neighborhood
1	137759.0	1798.2	3	-0.14
2	122228.1	1715.4	3	-0.67
3	284897.3	1975.6	4	0.30
4	152319.4	1709.9	3	0.39
5	436653.1	2642.1	4	1.45
...	...	...	...	...
296	452100.7	2498.8	5	0.87
297	145301.2	1623.4	3	-0.42
298	512709.9	2918.6	4	1.87
299	230342.5	1826.1	4	0.12
300	121893.4	1433.9	2	-0.93

Housing data cont.

- Consider running two really simple feature selection algorithms
- First is simple correlation (which we later refer to as a filter method)
- Second is multiple regression (which we later refer to as an embedded method)

Housing data cont.

- Consider running two really simple feature selection algorithms
- First is simple correlation (which we later refer to as a filter method)
- Second is multiple regression (which we later refer to as an embedded method)

Housing data cont.

- Consider running two really simple feature selection algorithms
- First is simple correlation (which we later refer to as a filter method)
- Second is multiple regression (which we later refer to as an embedded method)

Housing data results comp.

Feature	Pearson r	p-value	Significant ($\alpha = 0.05$)?
sqft	0.854	1.35×10^{-86}	✔ Yes
bedrooms	0.733	7.42×10^{-62}	✔ Yes
neighborhood	0.381	8.74×10^{-12}	✔ Yes

Figure: Housing data correlation

Feature	Coefficient	Std. Error	t-stat	p-value	Significant?
Intercept	7,771	8,845	0.88	0.380	✘ No
sqft	153.55	11.64	13.19	$< 1e-15$	✔ Yes
bedrooms	-2,920	4,388	-0.67	0.506	✘ No
neighborhood	21,190	1,863	11.38	$< 1e-15$	✔ Yes

Figure: Housing data regression

Observations/thoughts?

Feature	Pearson r	p-value	Significant ($\alpha = 0.05$)?
sqft	0.854	1.35×10^{-86}	✔ Yes
bedrooms	0.733	7.42×10^{-62}	✔ Yes
neighborhood	0.381	8.74×10^{-12}	✔ Yes

Figure: Housing data correlation

Feature	Coefficient	Std. Error	t-stat	p-value	Significant?
Intercept	7,771	8,845	0.88	0.380	✘ No
sqft	153.55	11.64	13.19	$< 1e-15$	✔ Yes
bedrooms	-2,920	4,388	-0.67	0.506	✘ No
neighborhood	21,190	1,863	11.38	$< 1e-15$	✔ Yes

Figure: Housing data regression

Question

In this housing data example, should we include the number of bedrooms as a feature?